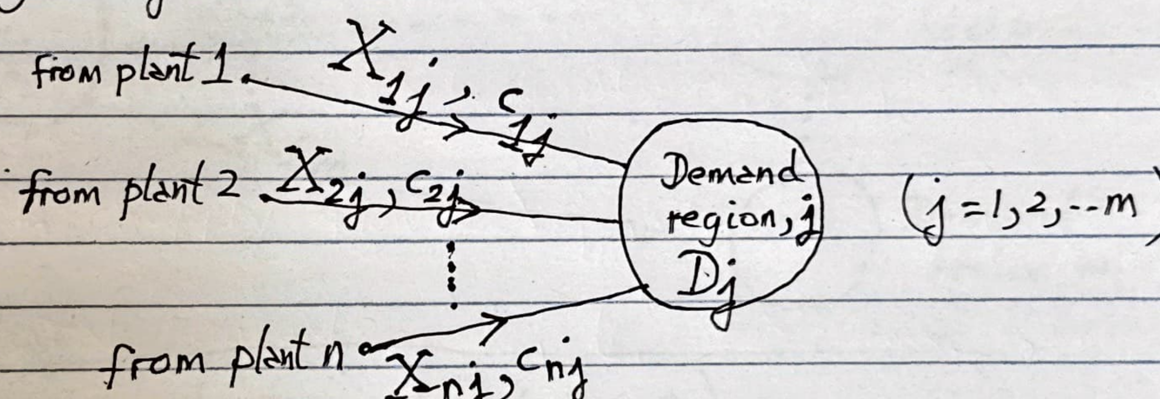
2. Determine the constraints

c(1) : Demand side constraints

For a given demand region, j, all incoming <u>supply</u> flowing into region j, must be greater than (or equal to) the demand for region j



For any demand region j, $(j = 1, 2, \cdots m)$

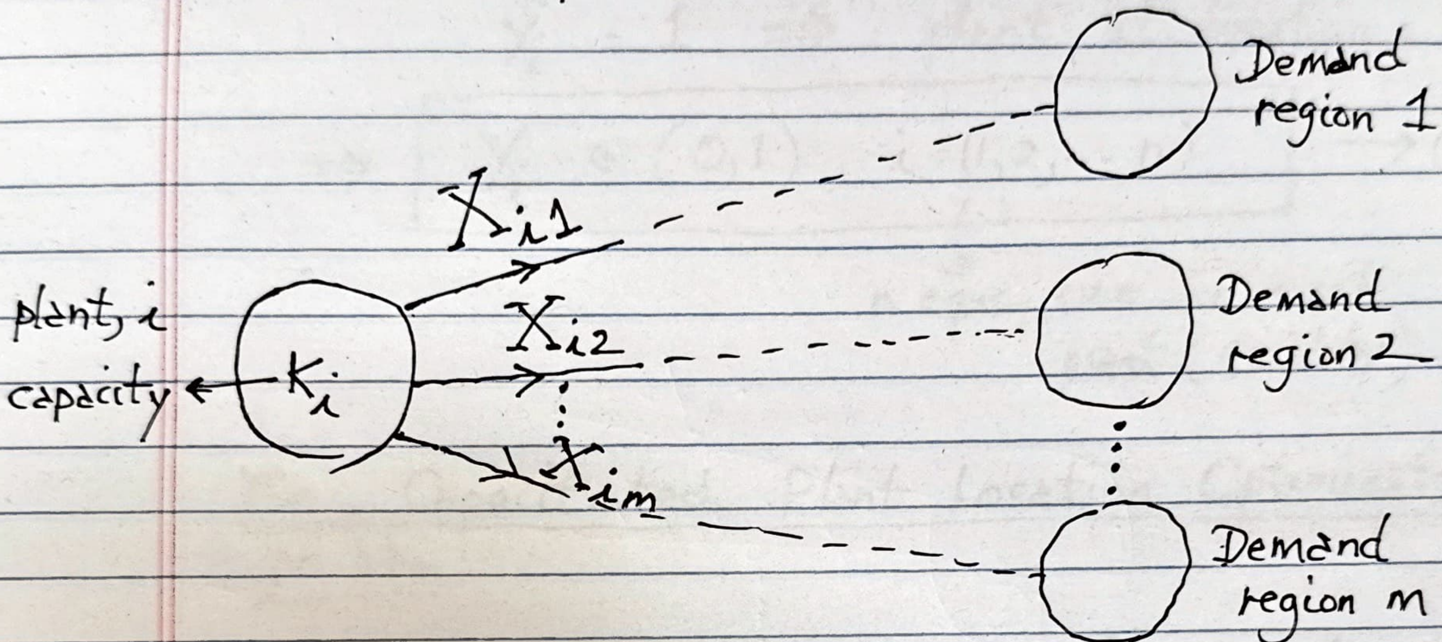
supply to region j, $X_{1j} + X_{2j} + \cdots X_{nj} \geq \underbrace{D_j}_{\text{demand for region } j}$

for each demand region, j

$$\boxed{\sum_{i=1}^{n} X_{ij} \geq D_j \; , (j = 1, 2, \cdots m)} \rightarrow (2)$$

"m" eqns., one for each region of demand c(2): supply side constraints

For each plant, i



$$\boxed{\begin{array}{l}\text{for each plant } i, (i=1,2,\dots n)\\ Y_i K_i \geq \sum_{j=1}^{m} X_{ij},\end{array}} \rightarrow (3)$$

$\underbrace{i=1,2,\dots n}$

n eqns, one for each plant, i

c(3) : Plant location constraint

$Y_i = 0 \Rightarrow$ no plant at location i

$Y_i = 1 \Rightarrow$ plant at location i

$$\Rightarrow \boxed{Y_i \in (0,1), \quad i = (1, 2, \dots n)} \rightarrow (4)$$

n eqns, one for each plant (facility)

The Capacitated Plant Location Optimization problem :

Determine the plant locations, Y_i, & plant capacities, K_i, & supply flows, X_{ij}, to minimize the total cost C, (objective), given by eqn. (1), subject to three sets of constraints, given by eqns. (2), (3), & (4)

Approach and Implementation

1. Set up the objective function, eqn. (1), and the constraints, eqns. (2), (3), and (4) in Microsoft Solver.
   (Please do the MS Solver tutorial on Canvas)

2. Use "MS Solver" to determine Y_i, K_i & X_{ij} s.

Important notes (for HW #7)

(i) HW #7, Prob. 5 ⇒ do the worked example in the "Network Design in the SC" chapter of the textbook

(ii) HW # 7, Prob. 6 ⇒ DRY ICE, which I've set up for you, above

(iii) do the MS Solver tutorial before attempting Probs. 5& 6 above.

TIM 172B, MOT - II (SCM), Lecture # 19
3/10/26

<u>Agenda</u>

0. Complete the Capacitated Plant Location Model derivation

1. HW # 7
   - Prob. 5 (Worked example) } discussed
   - Prob. 6 (Dry Ice) } above

2. HW # 7
   - Remaining problems -

2. HW # 7.

Prob. 7 : $\underbrace{\text{Books-on-Line}}_{\text{BOL}}$ (aka Amazon at inception)

- This problem is a great example of integration : includes inventory (cycle & safety), facilities, & transportation.

Objective : BOL needs to optimize (minimize) the total cost of its SC Network

The problem asks you to explore various options for locating warehouses in order to minimize total cost.

Process (or Plan or Approach or Procedure) :

1. Create a table of options for facility (warehouse) locations :

	Warehouse Location Scenarios		
	East	Central	West
1. Current Scenario	0	0	1
2. 1 Warehouse in Central Zone	0	1	0
3. 1 Warehouse in each Zone	1	1	1
4. Other ?		?	

2. For each of the above options (or scenarios) calculate the total cost

Total cost = Transportation (or shipment cost)
+ Cycle Inventory holding cost
+ Safety Inventory holding cost
+ fixed cost of the warehouse
+ variable cost of operating the warehouse

Notes:

(a) Transportation cost is dependent on the source (location of warehouse) & destination location $\Rightarrow$ specified as a weekly cost

(b) Warehouse cycle inventory holding cost is an annual cost $= \left(\frac{1}{2} Q_L\right) hC$

(c) Warehouse safety inventory holding cost is an annual cost $= (ss)(hC)$

safety stock

Replenishment Policy : Periodic Review (weekly)

$$T_R = 1 \text{ week}$$

Recall that, $SS = \left(\sqrt{T_R + L}\right)\left(\sigma_{L+T_R}\right) F^{-1}\left[(CSL)_{desired}\right]$

99.7 %
$\equiv 0.997$

(d) Fixed cost of the warehouse

This cost needs to be averaged over 10 years (or 520 weeks) $\Rightarrow$ weekly cost.

(e) Variable cost of operating the warehouse is given as a weekly cost

<u>Observation</u> : some of the costs are weekly & some are annual

Therefore, reduce all costs to a common weekly basis, i.e. per week.

Create a table of options, & select the option that minimizes total cost

options → / costs	1 current: (0 0 1)	2 (0 1 0)	3 (1 1 1)
ⓐ Transp ⓑ Cycle ⓒ .... ⓓ ... ⓔ	\$	\$....	\$...
Σ or total weekly cost	\$ ____	\$____	\$ ___

If you do the calculations correctly, then you will find that the <u>dominant cost</u> for this problem ~~is the~~ transportation cost, & you will discover the option that minimizes the total cost is option 3, (1,1,1), or "warehouse in each zone.

check on your calculations: The total cost for each option $\approx$ \$100,000 per wk